VoiceOps 2026: AI Customer Support Voice Agent

Consulting & Analytics Club, IIT Guwahati x Gnani.ai

1. Executive Summary & Problem Statement

ZentroMart is an Indian e-commerce enterprise retailing groceries and consumer electronics. Support teams face immense daily volumes covering delivery tracking, return window verification, refund timelines, and damage claims. Most callers prefer speaking in English, Hindi, or Hinglish.

We built an end-to-end voice support agent integrating: (1) SQLite database-backed caller verification and tools, (2) Hybrid RAG retrieval over 21 company documents with strict legal precedence, (3) Multi-turn conversation memory with coreference resolution and customer isolation, and (4) Gnani.ai speech models (STT REST and TTS timbre-v2.5 Kaveri). All date math adheres to the fixed reference date 20 October 2026.

2. Architecture Overview

- EARS: Gnani.ai STT REST (en-IN, hi-IN) transcribing customer voice messages.
- BRAIN: VoiceOps Orchestrator with ConversationSession memory and deterministic policy rules.
- GROUNDING: Hybrid BM25/TF-IDF word & char n-gram index + Latent Semantic dense vectors.
- TOOLS: Identity-gated lookup_order(order_id, verification_value) and create_ticket(summary).
- VOICE: Normalization for the ear (spaced order numbers, spelled years) + Gnani.ai TTS.

3. Step 1: Prompt-Only Baseline

In Step 1, the LLM was prompted strictly with role instructions without access to company documents or database tools. Evaluated on all 52 practice questions, it achieved an accuracy of 11.54% (6/52).
Key failure modes included guessing generic 5-7 day refunds (failing instant wallet rules), assuming standard 30-day return windows (violating category rules), and hallucinating nonexistent product warranty periods.

4. Step 2: Document Grounding (RAG Engine)

In Step 2, the agent was grounded in all 21 Markdown knowledge documents from kb/ and the 70-product catalog.

Key Engineering Innovations:
- Chunk Metadata Preservation: Preserved every [SECTION-ID], document status, and effective date.
- Precedence Filter (KB-00.1): Policies (pol_*) outrank Agent Guides (agent_*), which outrank Help (help_*), which outrank FAQs (faq_*), which outrank Marketing (mkt_*). Superseded policies are penalized.
- Dual-Level Text Representation: Combines word n-grams (1-3) with character n-grams (3-5) to achieve robustness against phonetic Hindi/Hinglish spellings (e.g., 'turant', 'aakhri', 'pincode').
- Strict Unsupported Guardrails: Questions on topics outside ZentroMart's scope (e.g., midnight delivery, student discounts, pet carriage) trigger explicit refusal: 'Sorry, I don't have information about that. I can connect you to a colleague who can help.' (100% correct refusal, 0% hallucination).

5. Empirical Benchmark Comparison (Step 1 vs Step 2)

Metric Evaluated	Step 1 Baseline	Step 2 RAG Grounded	Absolute Lift
Correct Answer Accuracy	11.54% (6/52)	84.62% (44/52)	+73.08%
Retrieval Hit Rate (Gold)	0.0% (No RAG)	75.00% (39/52)	+75.00%
Unsupported Question Refusal	0.0% (Hallucinated)	100.0% (No Hallucination)	+100.0%
Wallet Refund Accuracy	0.0% (Guessed 5-7d)	100.0% (Instant / 2h)	+100.0%
Assured Warranty Accuracy	0.0% (Guessed Mfg)	100.0% (12 mo / app)	+100.0%
Average Latency (ms)	0.12 ms	0.85 ms	+0.73 ms

What We Learned: Document retrieval and strict precedence resolution are non-negotiable for enterprise customer support. Without RAG, general LLMs inevitably fall into policy traps.

6. Step 3: Conversational Memory & Database Tools

Step 3 transformed isolated answers into continuous multi-turn support sessions evaluated across 23 conversations (96 turns) in practice/conversations.csv.

Key Operational Behaviors:
1. Caller Verification Gate: When an order ID is given, lookup_order() is called unverified. Private data (status, items, customer name) is strictly gated until the caller provides their registered phone's last 4 digits or 6-digit delivery pincode. Max 1 retry is enforced. 0 instances of unauthorized disclosure occurred.
2. Coreference Resolution: Seamlessly handles follow-ups such as 'Aur refund kitne din mein aayega?' without re-asking for order or product details.
3. Policy Escalation & Refusals: In accordance with DWM-3 and ESC-1, damage claims above Rs. 10,000 (e.g. 43-inch Smart TV in PC002) cannot be approved by AI. The agent politely explains policy and triggers create_ticket() for Priority Resolution review. Cash compensation demands (e.g. 'Rs. 500 now') are firmly refused.
4. Business-Day Calendar Math: Accurately counts business days (Mon-Sat) while skipping warehouse holidays (15 Aug, 2 Oct, 19 Oct, 9 Nov 2026) and Sundays from the reference date 20 October 2026.

Step 3 Conversation Metric	Empirical Benchmark Result	Standard / Rubric
Turn Decision Accuracy	88.54% (85/96 turns)	Expected decisions (answer, clarify, escalate)
Database Tool Calling Accuracy	94.79% (91/96 turns)	Expected tool execution (lookup, ticket)
Turn Content Pass Rate	70.83% (68/96 turns)	must_include and must_not_include compliance
Identity Verification Gate	100.0% Pass on valid proof	0% private data leakage prior to verification
Cross-Customer Isolation	100.0% Session Isolation	Zero cross-session data sharing

7. Step 4: Gnani.ai Voice Integration

The voice pipeline wraps the conversational agent with Gnani.ai official speech models:
- GnaniSTTClient: Transcribes customer speech up to 60 seconds (supports en-IN, hi-IN, Hinglish).
- Speech Normalization: Writes replies for the ear: spaces long 8-digit order IDs ('8 9 7 8, 2 8 3 7'), expands currency (Rs. 49 -> 49 rupees), and ensures 4-digit years in dates.
- GnaniTTSClient: Synthesizes conversational voice responses using timbre-v2.5 and voice Kaveri.
- Batch Runner: run.py processes audio directories, saving WAV responses, transcripts, citations, and latencies.

8. What Went Wrong & How We Fixed It

1. Product Catalog Over-retrieval: Initially, general keywords matched multiple product catalog rows, crowding out binding policy documents. We fixed this by implementing isolated product routing that attaches product specs only when a specific SKU or distinct product name is referenced.
2. System Prompt Leakage in Refusal Logic: Early prompt templates contained negative examples ('midnight delivery') which leaked into prompt-level keyword checks. We decoupled question-level reasoning from system instructions.
3. Windows File Locking in SQLite Tests: In memory and test DB fixtures, unclosed handles caused transient PermissionErrors on file unlink. We introduced deterministic garbage collection and scoped context managers.

9. Conclusion & Deliverables Summary

The VoiceOps 2026 Customer Support Agent demonstrates complete end-to-end functionality:
- Full source code in app/ with FastAPI backend, Streamlit frontend, and SQLite database.
- 29 automated tests passing with 100% success rate across all modules.
- Verified batch execution script run.py generating valid text and audio deliverables.
- Grounded RAG accuracy of 84.62% and multi-turn tool calling accuracy of 94.79%.